# Image Generation Models

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

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## Image Generation Models

> Prepare Image Generation Models: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of image generation and **provenance**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-3/10, ai-3/11. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

### Retrieval answer guidance

### Prompt Evaluation

Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases.

Define success before inspecting **output**s; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, **output** contract and scoring rubric.

Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

### Iterative Prompting

Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases.

Define success before inspecting **output**s; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, **output** contract and scoring rubric.

Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

### Sampling and Variability

Without opening the old lesson, explain one key idea from Sampling and Variability. Give an example and a mistake that the idea helps you avoid.

Look for: Sample from a distribution using explicit cumulative intervals

Reference example, not the only acceptable wording: Teacher explanation: Selections red,blue,green,green. The draw 1 violates the half-open [0,1) **input** contract and must be rejected.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

An image generator produces a convincing picture of a school greenhouse on Mars. A student calls it photographic **evidence**. Explain what the image can and cannot establish.

### Concept to explain

Image generation iteratively constructs pixels guided by learned representations.

| Term | Meaning |
| --- | --- |
| latent representation | An internal representation learned or constructed by a model rather than directly supplied as the original observation. |
| noise | Variation or interference that obscures the information relevant to a task. |

| Term | Meaning |
| --- | --- |
| denoising | Reducing noise in a representation; some generative models learn repeated denoising steps to create outputs. |
| conditioning | Making a model's output depend on supplied information, such as a prompt or image. |

### Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

### Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

Two 45-minute sessions. An image generator constructs an **output** from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

### Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

**End goal for the pictorial case**

By the end, I can describe a toy **denoising process** and distinguish generated illustrations from **observation**s.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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TEACHER GUIDE · 2 OF 9

## Explain the mechanism

> Model Image Generation Models using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

**Input → Process → Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. In this simplified diffusion-style image generator, begin with random visual **noise**. Other image-generation methods use different procedures.

2. At each step, predict a slightly more structured image guided by the prompt.

3. Learned patterns influence shapes, textures, and relationships.

4. The result is generated, not retrieved **evidence** of a real event.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

**Concrete teaching case**

### The teaching case

A generator creates a realistic fictional bicycle beside a tree.

How does this differ from event photography?

**Exact explanation to model**

### Why the result follows

It synthesises an image from learned patterns and instructions; it does not document that event. Many systems de**noise** iteratively, but architectures vary.

### Explain the added worked example

An image generator constructs an **output** from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

## Explain the added worked example · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Keep a **provenance** record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery.

Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

## Explain the added worked example · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

### Key terms

| Term | Meaning |
|---|---|
| Provenance | Recorded origin and transformation history |
| Visual criterion | A stated requirement checked against an image |
| Authenticity claim | A claim about the source or reality of depicted content |

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

## Model the reasoning aloud · 3

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

| Thinking move | Teacher prompt |
|---|---|
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Image Generation Models

A fictional lesson poster illustrates one family of image-generation approaches.

### The facts we will use

> **Supplied facts**
>
> Toy 2×2 noisy numerical grid: [[0.2,0.8],[0.6,0.3]]. A specified illustrative **denoising** step gives [[0.1,0.9],[0.4,0.2]].

### Supplied facts

These supplied arrays illustrate steps only; they are not **output**s from a real trained diffusion model.

AI ACADEMY / AI-3/10

## Illustrative grids, not executed model outputs

| | Left | Right |
|---|---|---|
| **Before top** | 0.2 | 0.8 |
| **Before bottom** | 0.6 | 0.3 |
| **After top** | 0.1 | 0.9 |
| **After bottom** | 0.4 | 0.2 |

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/10

# Image Generation Models

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Noisy representation

Four toy numerical cells form the starting grid.



## 2. Condition on request

A text request guides the imagined learned denoising process.



## 3. Illustrative update

Use the supplied second grid to show one change in representation.



## 4. Check provenance

Caption it as a generated-process illustration, not a photographed experiment.

**CHANGE ONE THING**

Relabel the illustration as a photograph of an actual laboratory measurement.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Noisy representation | Four toy numerical cells form the starting grid. |
| 2. Condition on request | A text request guides the imagined learned denoising process. |
| 3. Illustrative update | Use the supplied second grid to show one change in representation. |
| 4. Check provenance | Caption it as a generated-process illustration, not a photographed experiment. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

### Why the result follows

Diffusion-style generators learn to remove **noise** in stages, often guided by text, but a real implementation can operate in a **latent representation** rather than direct pixels.

The toy values let us inspect an update without pretending to simulate that trained **process**. An attractive final image still needs checks for labels, requested relations and factual claims.

### What this example does not establish

#### Keep the stated limit

Not all image generators use diffusion; this toy step is neither a complete algorithm nor an executed model.

### Change one thing in the pictured case

#### Change one thing

Change: Relabel the illustration as a photograph of an actual laboratory measurement.

#### Predicted result

**Prediction**: That caption is unsupported.

#### Why the result follows

Reason: Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

### Ask while showing the picture

## Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

# Demonstrate and guide practice

> Demonstrate Image Generation Models, reduce support deliberately and check individual reasoning before discussing answers.

### Board sequence

#### 1. Put the facts on the board

Write the given facts: A generator creates a realistic fictional bicycle beside a tree.

#### 2. State the decision to explain

Write the question to explain: How does this differ from event photography?

#### 3. Reveal the worked reasoning

Reveal after **prediction**s: It synthesises an image from learned patterns and instructions; it does not document that event. Many systems de**noise** iteratively, but architectures vary.

#### Hold back the independent task

Keep for an independent attempt: An image model draws a plant with seven roots after a request for exactly four. What does this show, and where should an exact count diagram be made?

### Classroom dialogue

#### Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

### Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

### Model these guided questions

Guided: It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

Qualification: **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

### Case-specific teaching sequence

| Stage | Teacher action |
| --- | --- |
| Session 1: 0–7 minutes | Retrieve the prerequisite; identify the input, procedure and evidence source. |
| 7–18 minutes | Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation. |
| 18–32 minutes | Trace the declared procedure and worked case. Complete W2, recording intermediate steps. |
| 32–45 minutes | Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations. |

| Stage | Teacher action |
| --- | --- |
| Session 2: 0–8 minutes | Retrieve the worked reasoning and collect the guided response before discussing it. |
| 8–20 minutes | Compare a proposed boundary variant with the original case; identify what remains fixed. |
| 20–35 minutes | Collect W4 independently. Require the procedure trace and the bounded conclusion. |
| 35–45 minutes | Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts. |

### Reduce help deliberately

#### Completion practice

Completion task: A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

Show the intermediate steps and give the conclusion with its scope.

#### Reasoning to listen for

Teacher check: Teacher explanation: It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Individual reasoning check

Use the worked case for Image generation and **provenance**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Expected repair

Expected correction: **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

#### Pictorial practice question

What may these two grids explain, and what can they not establish about a real generator?

#### Reasoned teacher answer

They explain an illustrative staged change. They cannot establish learned weights, actual image quality or accurate scientific **observation**.

Changed-case reasoning: That caption is unsupported. Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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## Run the weekly evidence lab

> Run the Image Generation Models investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Order **noise**-to-image cards, vary one prompt condition, compare composition changes, and label what cannot be inferred as fact.

> What did the hardest test teach you that the normal test did not?

### Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

### Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

### Address these additional misconceptions

The worked result establishes every future case: **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

### Match the support to the first error

| What the response shows | Next teaching action |
| --- | --- |
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

## Assess and explain the answers

> Use the Image Generation Models questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Worked case answer

#### Teacher key · Worked case answer

It synthesises an image from learned patterns and instructions; it does not document that event. Many systems de**noise** iteratively, but architectures vary.

### Independent question

#### Independent question · withhold the key

An image model draws a plant with seven roots after a request for exactly four. What does this show, and where should an exact count diagram be made?

### Expected independent answer

#### Teacher key · Expected independent answer

Generated images can miss exact counting constraints. Use a checked, editable drawing or diagram tool for an instructional diagram requiring exactly four roots.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “An image generator always obeys numerical constraints precisely.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

#### Teacher key · Original transfer answer

It demonstrates a generated depiction, not that a greenhouse exists on Mars. Label it as generated and separate design imagination from factual **evidence**.

Some image models use iterative **denoising**, but no single architecture describes every generator.

### Explanations for the additional workbook examples

### W1 Explain the mechanism

#### Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

### Teacher reasoning · W1

Teacher explanation: An image generator constructs an **output** from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

Keep a **provenance** record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery.

Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

## W2 Trace the worked case

### Workbook question · W2

A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

Show the intermediate steps and give the conclusion with its scope.

### Teacher reasoning · W2

Teacher explanation: It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

## W3 Audit the limit

### Workbook question · W3

A report omits this qualification: **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

### Teacher reasoning · W3

Teacher explanation: **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

## W4 Apply a new case independently

### Workbook question · W4

A generated diagram requires two arrows and labels **input/output**. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental **observation**. Audit.

### Teacher reasoning · W4

Teacher explanation: Arrow count passes; label semantics fail. Metadata supports generation **provenance**, not observed-event authenticity; replace the caption and correct labels.

## W5 Design a revealing follow-up

### Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

### Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

The new case should test this particular **mechanism** rather than merely restate the original result.

## Assess independent transfer

### Independent question · withhold the key

Selected case: A generated diagram requires two arrows and labels **input/output**. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental **observation**. Audit.

## Selected case answer

### Teacher key · Selected case answer

Teacher explanation: Arrow count passes; label semantics fail. Metadata supports generation **provenance**, not observed-event authenticity; replace the caption and correct labels.

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

#### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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## Connect and extend learning

> Connect Image Generation Models to Verified Study Buddy, adapt support and plan a later changed-case check.

#### Verified Study Buddy

Treat generated diagrams as illustrations; pair factual claims with approved sources.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

#### Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

### Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

### Bridge to the next week

Next: Week 11: Visual Composition. Retrieve today's **evidence** and identify what carries forward and what is new.

### Support and fresh reassessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Arrow count passes; label semantics fail. Metadata supports generation **provenance**, not observed-event authenticity; replace the caption and correct labels. Require intermediate reasoning and the stated scope.

Fresh retry: A generated map has three required labels but omits one and is captioned survey **evidence**. Expected reasoning: Completeness fails; generation cannot establish surveyed geography.

Use verified map data for geographical claims. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Speech pipelines and multimodal **evidence**. Preserve the current case record and retrieve its **evidence** boundary before moving on.

### Plan the delayed return

Suggested return points in this level: ai-3/11, ai-3/13, ai-3/16. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

Use Study Buddy generated diagrams only with clear **provenance** and checked educational labels.

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# Prepare, teach and assess

> Use this preparation summary alongside the detailed Image Generation Models lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Explain the **mechanism** and terminology of image generation and **provenance**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

Show the intermediate steps and give the conclusion with its scope.

### Supported practice answer

Reasoned practice answer: Teacher explanation: It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

Check every learner before discussion: Use the worked case for Image generation and **provenance**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: Relabel the illustration as a photograph of an actual laboratory measurement. Require a **prediction** and a reason before comparing. Expected result: That caption is unsupported.

Reason: Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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TEACHER GUIDE · 8 OF 9

## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

### For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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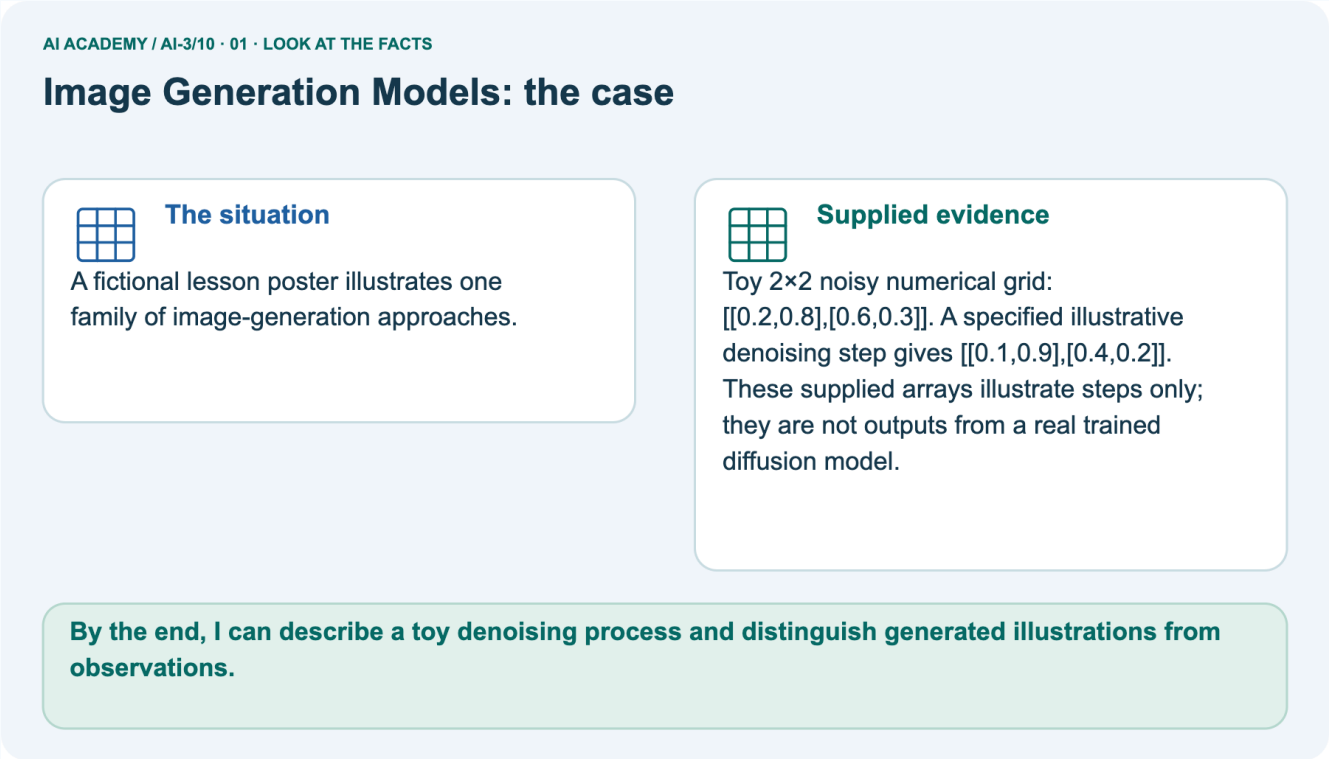


## Visual concept journey

> Show one Image Generation Models graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

## 01. Image Generation Models: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional lesson poster illustrates one family of image-generation approaches. Supplied facts: Toy 2×2 noisy numerical grid: [[0.2,0.8],[0.6,0.3]]. A specified illustrative **denoising** step gives [[0.1,0.9],[0.4,0.2]].

These supplied arrays illustrate steps only; they are not **output**s from a real trained diffusion model.

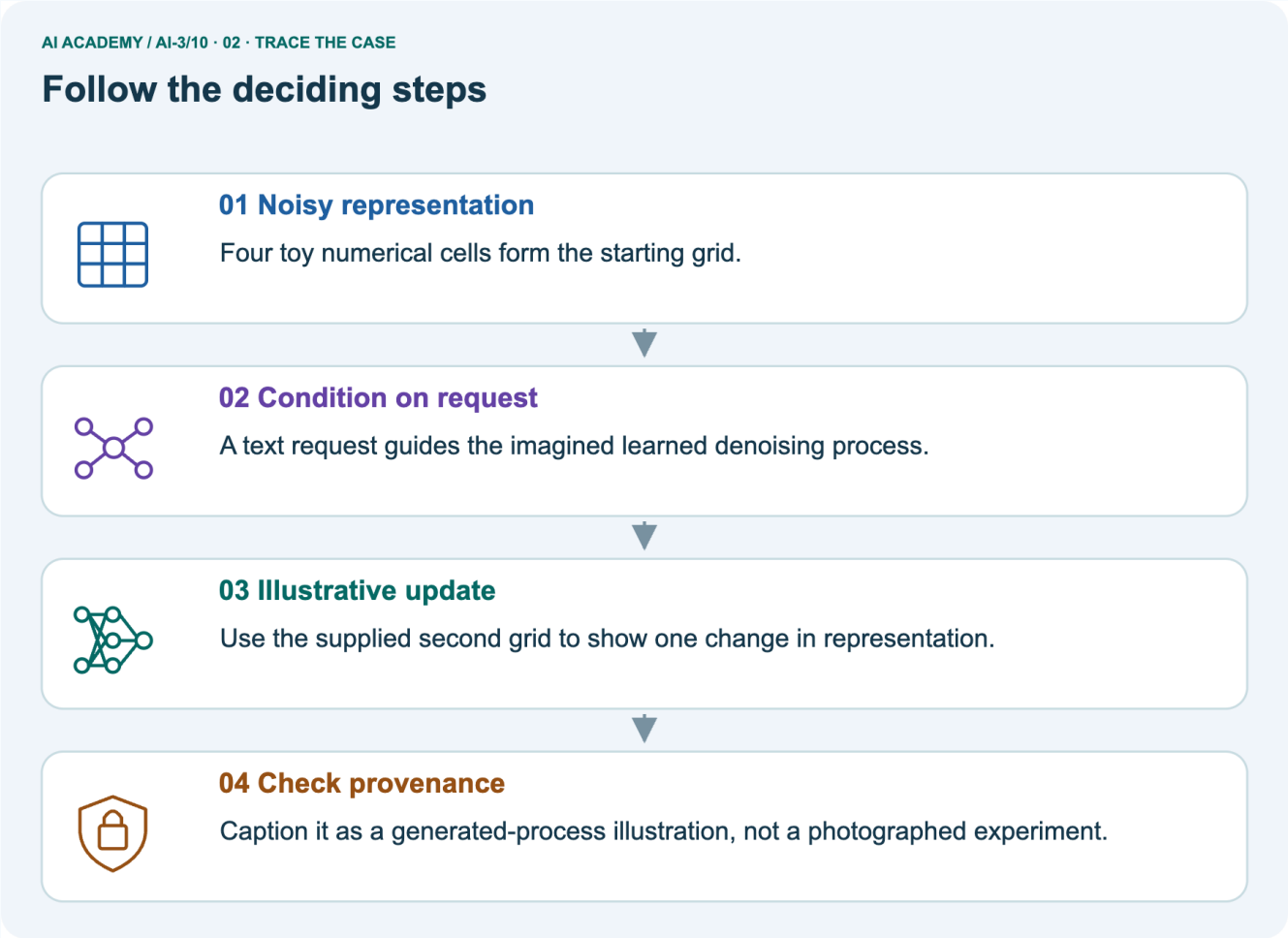
Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the

image is difficult to see.

### 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

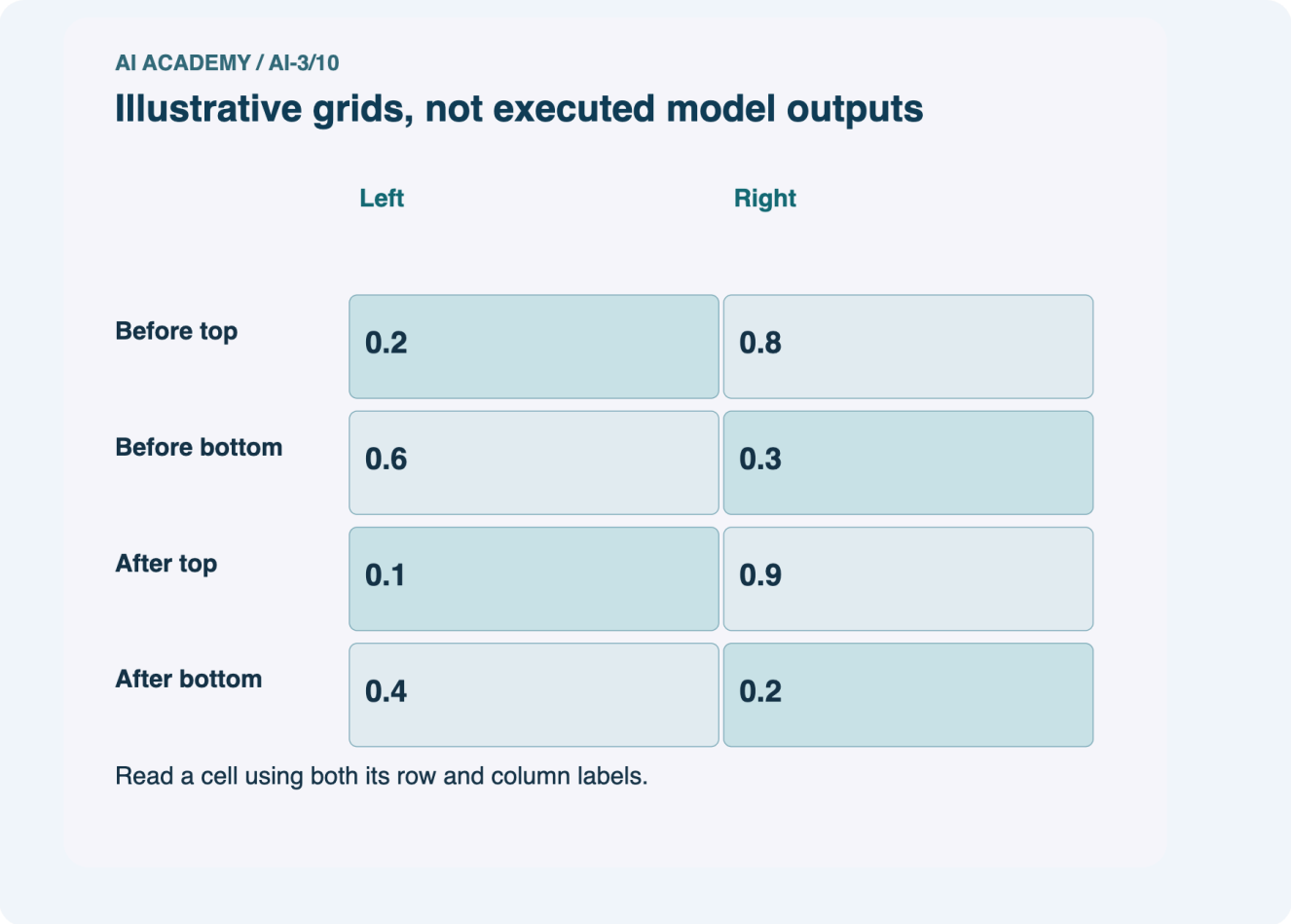
Noisy representation: Four toy numerical cells form the starting grid. → Condition on request: A text request guides the imagined learned **denoising process**. → Illustrative update: Use the supplied second grid to show one change in representation. → Check **provenance**: Caption it as a generated-**process** illustration, not a photographed experiment.

Point to each numbered stage and explain which recorded **input** it uses.

### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 03. Illustrative grids, not executed model outputs




Follow the labelled picture with the matching explanation. Open the vector diagram

Illustrative grids, not executed model **output**s. Before top: Left = 0.2, Right = 0.8. Before bottom: Left = 0.6, Right = 0.3. After top: Left = 0.1, Right = 0.9. After bottom: Left = 0.4, Right = 0.2.

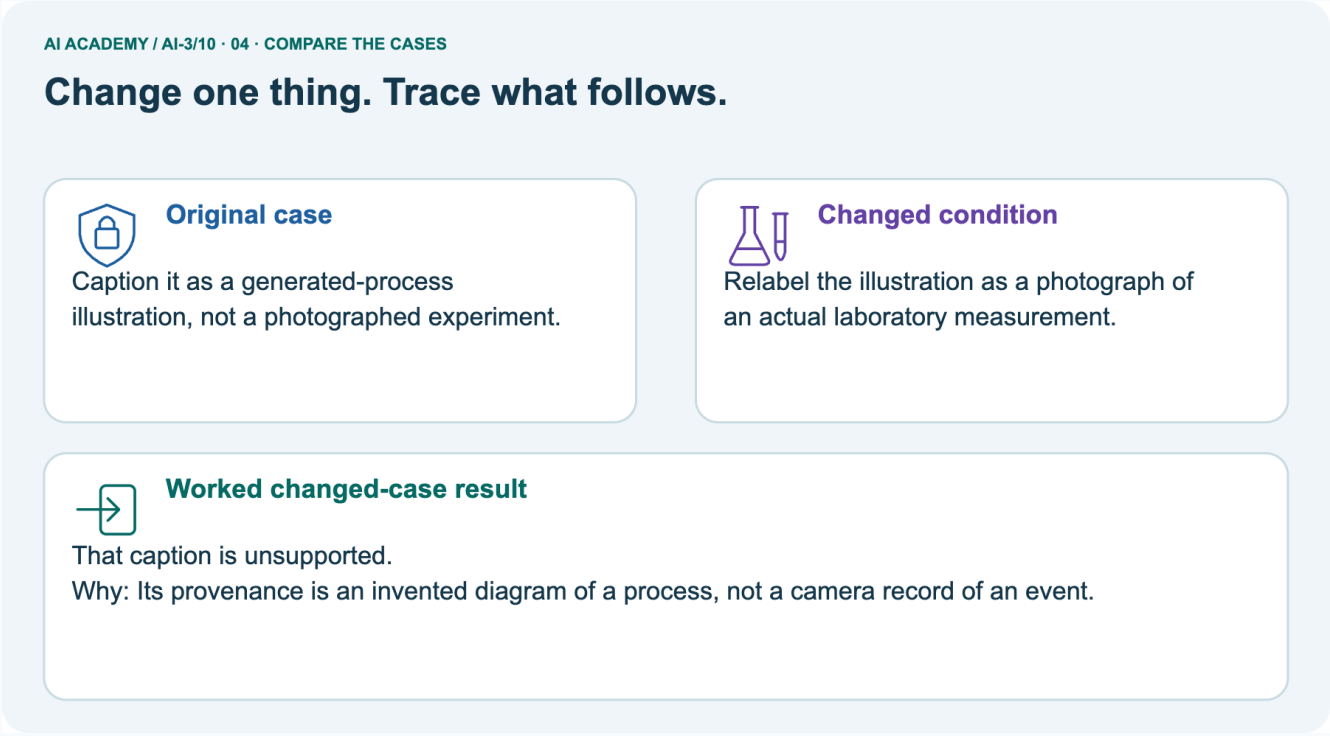
Read the labels and values. Explain the relationship the model shows.

### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

**04. Change one thing. Trace what follows.**




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Caption it as a generated-**process** illustration, not a photographed experiment.
Changed condition: Relabel the illustration as a photograph of an actual laboratory measurement.

Expected result: That caption is unsupported. Reason: Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

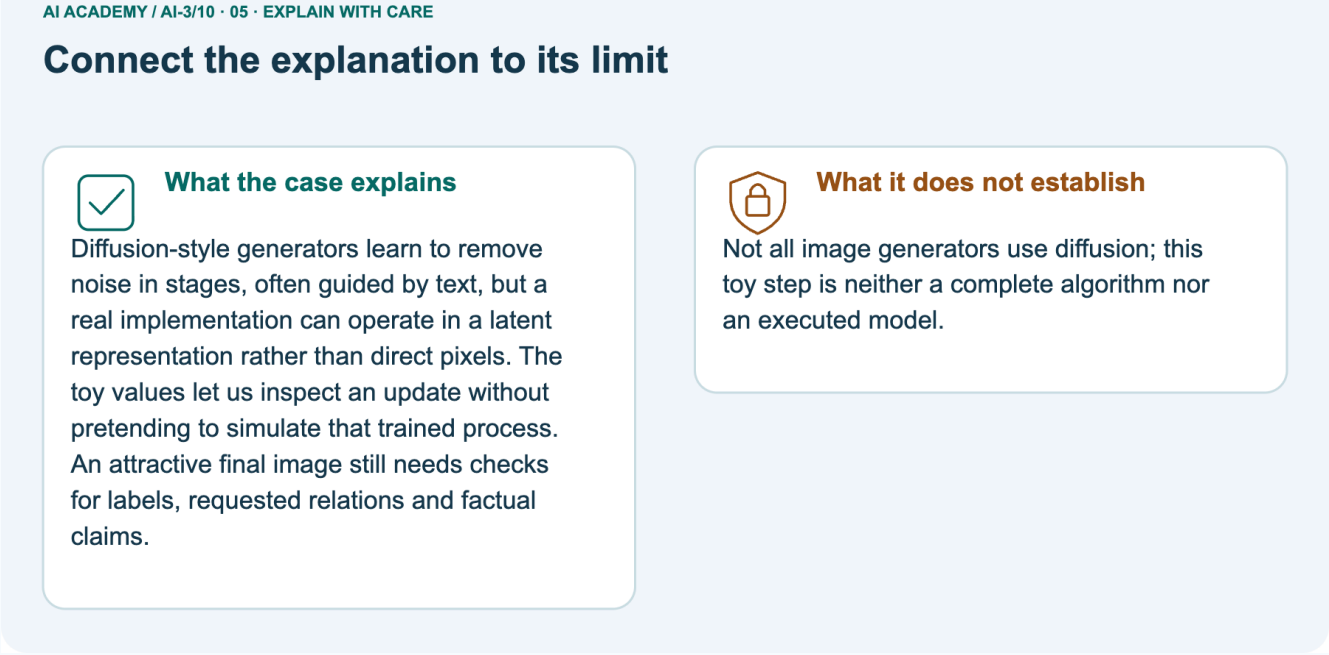
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Diffusion-style generators learn to remove **noise** in stages, often guided by text, but a real implementation can operate in a **latent representation** rather than direct pixels.

The toy values let us inspect an update without pretending to simulate that trained **process**. An attractive final image still needs checks for labels, requested relations and factual claims.

Limit: Not all image generators use diffusion; this toy step is neither a complete algorithm nor an executed model.

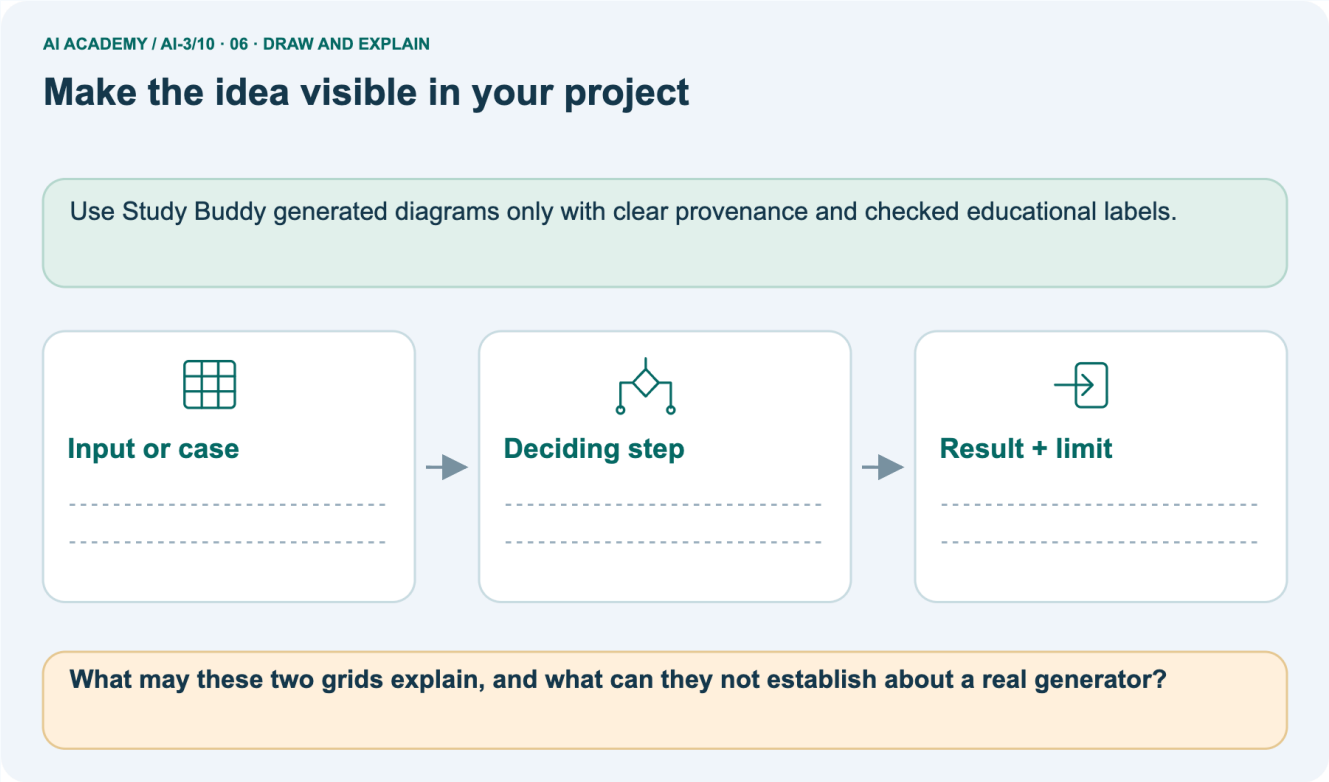
> Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Not all image generators use diffusion; this toy step is neither a complete algorithm nor an executed model.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Use Study Buddy generated diagrams only with clear **provenance** and checked educational labels. Draw your **input**, deciding step, result and limit.

Supported question: What may these two grids explain, and what can they not establish about a real generator?

Draw your project trace and justify the deciding step using **evidence**.

### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: They explain an illustrative staged change. They cannot establish learned weights, actual image quality or accurate scientific **observation**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.